

Q1 (2022 T1 Final Exam Q4b)

Calculate the current $i_0(t)$ in the circuit of Figure 1.

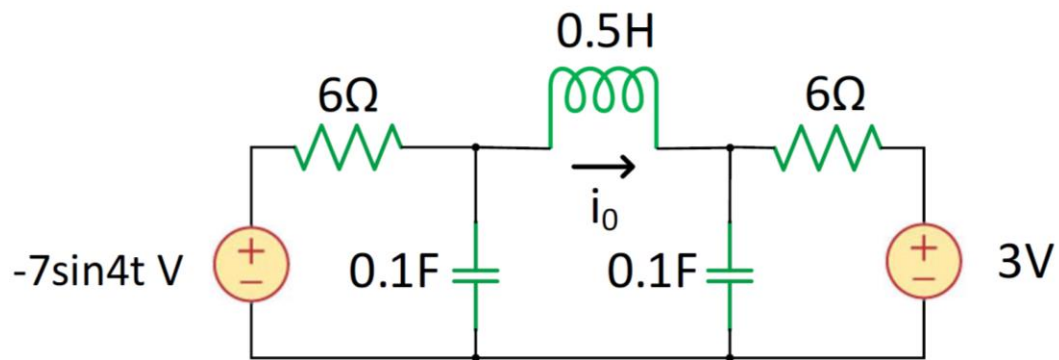


Figure 1

Answer: $i_0(t) = -0.25 + 0.94 \cos(4t + 74.39^\circ) \text{ A}$

Q2 (Modified from 2020 T1 Final Exam Q4b)

For the circuit shown in Figure 2, find the Thevenin equivalent circuit as seen from the load impedance Z_L .

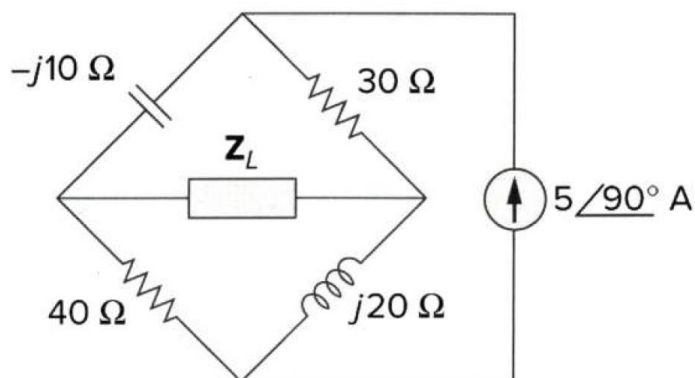


Figure 2

Answers: $Z_{th} = 20 \Omega$, $V_{th} = 70.84 \angle 81.84^\circ \text{ V}$.

Note: Since AC power (Topic 9) has not yet been introduced, the question has been slightly modified to exclude the power calculation. The solution for the modified question is included as part of the exam solution. You can focus on understanding how to calculate V_{th} (Thevenin equivalent voltage) and Z_{th} (Thevenin equivalent impedance) without delving into power calculations at this stage.

Q3 (Modified from 2023 T1 Final Exam Q5b)

The variable resistor in the circuit of Figure 3 is adjusted until it absorbs the maximum average power.

Find the Thevenin equivalent circuit as seen from the variable resistor R_{load} .

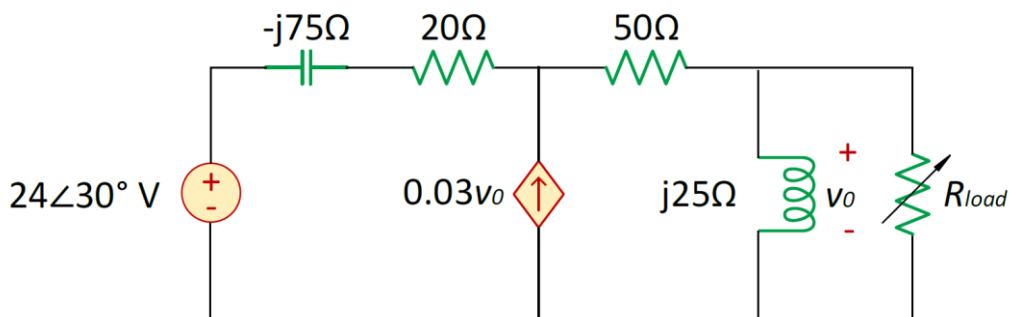


Figure 3

Answers: $Z_{th} = 38.79 \angle 123.14^\circ \Omega$, $V_{th} = 9.077 \angle -159.89^\circ \text{ V}$.

Note: Since AC power (Topic 9) has not yet been introduced, the question has been slightly modified to exclude the power calculation. The solution for the modified question is included as part of the exam solution. You can focus on understanding how to calculate V_{th} (Thevenin equivalent voltage) and Z_{th} (Thevenin equivalent impedance) without delving into power calculations at this stage.

Q4 (Modified from 2022 T1 Final Exam Q3b)

The circuit in Figure 4 represents an audio mixer, where v_1 and v_2 correspond to individual waveforms from the drums and guitar, respectively. If v_1 is a zero-phase sinusoidal signal with amplitude 240mV, v_2 is a zero-phase sinusoidal signal with amplitude 210mV, and the frequency is 60 Hz,

- Derive an expression for the combined phasor voltage V_0 in terms of the input voltages V_1 and V_2 (i.e., DO NOT substitute the values of V_1 and V_2).
- Substitute the values of V_1 and V_2 in your calculated V_0 from part i) and derive the combined waveform $v_0(t)$.
- Calculate the voltage across resistor R_1 (i.e., 12k Ω resistor) as a phasor.

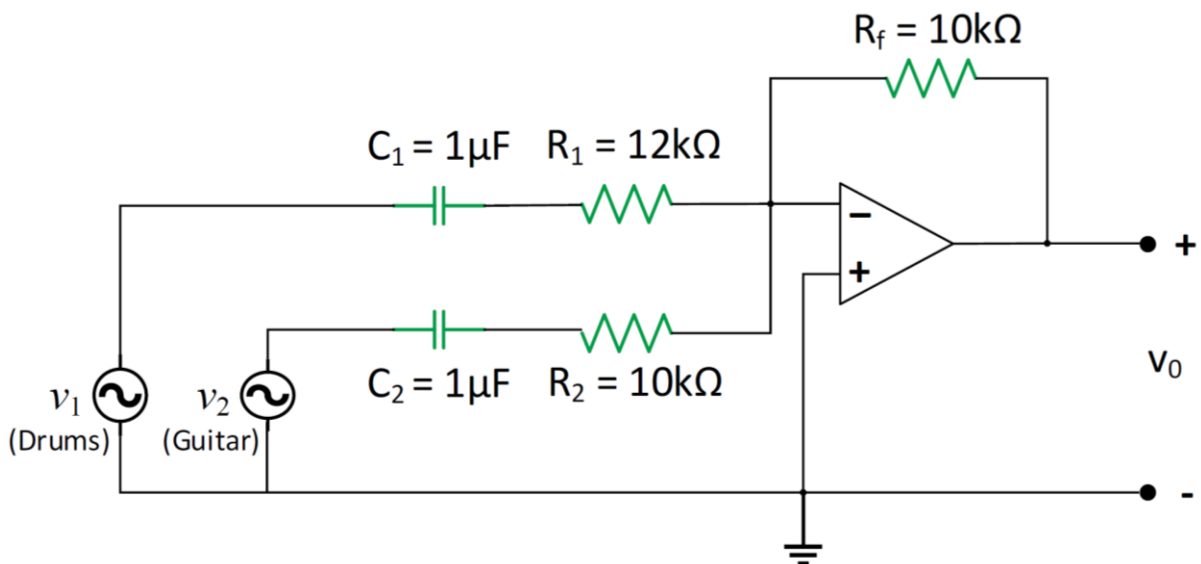


Figure 4

Answers:

- $V_0 = -(0.814 \angle 12.465^\circ V_1 + 0.967 \angle 14.86^\circ V_2) V$
- $v_0(t) = 0.4 \cos(120\pi t - 166.3^\circ) V$
- $V_{R1} = 0.234 \angle 12.465^\circ V$

Note: Since AC power (Topic 9) has not yet been introduced, the question has been slightly modified to exclude the power calculation. The solution for the modified question is included as part of the exam solution.